

Base plan P^*

strongest / highest cost

Retrieve $k = 16$

hybrid BM25 + dense

Rerank

cross-encoder

Generate

qwen-plus

Repair / NLI

entailment check

Rewrite lattice (cost $\downarrow$)

HybridQA calibration vouchers

R1 prefilter pushdown

$\hat{\epsilon} = 0.026$

retrieve $\rightarrow$ filter $\rightarrow$ rerank $\rightarrow$ gen $\rightarrow$ NLI

R2 truncate k

$\hat{\epsilon} = 0.081$

retrieve $k=8 \rightarrow$ rerank $\rightarrow$ gen $\rightarrow$ NLI

R3 elide rerank

$\hat{\epsilon} = 0.078$

retrieve $\rightarrow$ gen $\rightarrow$ NLI

R4 downgrade generator

$\hat{\epsilon} = 0.157$

retrieve $\rightarrow$ rerank $\rightarrow$ qwen-flash $\rightarrow$ NLI

compose along a path: telescoping voucher (Prop. 1)
union bound = search heuristic only

Division of labor

mirrors classical cost-based optimizers

Vouchers

ORDER the rewrite lattice
(may be loose / optimistic)

Certification

alone GUARANTEES the
deployed policy ($\text{FWER} \leq \delta$)

Assumption NSH

measured on real workloads;
not required for telescoping

Like a cost model: wrong estimates hurt optimality, never soundness.